CUSTOMER OPERATIONS COMPANION

Restate on Amazon EKS

Manual Deployment Reference
and Best Practices

PURPOSE
A field-ready companion for the cloud or platform engineer asked to install Restate into an existing EKS cluster. It condenses the repository runbook into deployment gates, commands, evidence, and safety boundaries.

Repository baseline: **b31a2e82**
Restate image: **1.7.7** Operator chart: **3.0.1**
Reference topology: **3 nodes / 48 partitions / replication 2**
Prepared: **4 September 2026**

01 / ORIENTATION

What this reference deploys

Restate is a durable execution runtime. This reference installs the runtime and its Kubernetes operator into an existing EKS cluster; it does not create the EKS cluster or deploy a customer application.

Layer	What it means	Ownership
EKS cluster	Existing AWS and Kubernetes infrastructure	Customer platform team; this repository does not create it
Restate cluster	Three stateful Restate server pods managed by a RestateCluster custom resource	Restate operator inside EKS
SDK service	Application code built with a Restate SDK and listening on port 9080	Application team; deployed separately

Deployment shape

Component	Reference value	Operational meaning
Restate	3 replicas; 24 vCPU and 50 GiB each	Requires three different eligible nodes
Data	256 GiB encrypted gp3 per pod; Retain	Monitor /restate-data and expand with operational headroom
Snapshots	Dedicated S3 bucket through IRSA	One bucket or unique prefix per Restate cluster
Network	Default-deny policies where enforced	Admin 9070 stays private; SDK port 9080 is isolated
Exposure	ClusterIP only	No ingress, DNS, public endpoint, or auth gateway is created

This guide is for

A cloud or platform engineer with AWS, EKS, IAM, Kubernetes, Helm, and change-management access. Detailed Restate internals are not assumed.

The successful handoff

A Ready three-node Restate cluster, Bound EBS volumes, a proven S3 snapshot path, verified network posture, and recorded ownership. No application or public ingress is implied.

CHOOSE ONE DELIVERY PATH

This artifact covers the manual AWS CLI, eksctl, Helm, and kubectl path. Please use either the manual or Terraform path for an installation. If Terraform will take over manually created resources, import them into state before switching.

02 / READINESS

Readiness checks before you begin

Please confirm each item below before the first apply and record the evidence in the change ticket. If an item is not yet confirmed, pause here and resolve it before continuing.

Check	Ready when	Why it matters
Identity	AWS account, region, EKS name, and kubectl context are confirmed	Prevents a correct deployment to the wrong environment
Capacity	3 eligible nodes each have at least 24 CPU and 50 GiB remaining by requests	Hard host anti-affinity allows one Restate pod per node
Storage	EBS CSI controller is healthy; persistent EBS is supported	The three data volumes must survive node replacement
IP family	EKS reports ipv4 and a Service IPv4 CIDR	The supplied Service-CIDR policy is IPv4-only
NetworkPolicy	Enforcement state and trust decision are documented	Objects are inert unless the CNI enforces them
Snapshots	A unique, dedicated S3 bucket exists in the target region	The configured snapshot prefix is shared by every default install
Authorization	Required AWS permissions and six kubectl can-i checks pass	Installation creates IAM, cluster-scoped RBAC, CRDs, and storage
IRSA	OIDC exists or the approved change includes creating it	The Restate ServiceAccount needs STS web identity

Set and verify the target

```
export CLUSTER=...
export REGION=...
export BUCKET=...   # globally unique; dedicated to this Restate cluster
export ACCOUNT="$(aws sts get-caller-identity --query Account --output text)"

aws sts get-caller-identity
aws eks describe-cluster --name "$CLUSTER" --region "$REGION" \
  --query \
  'cluster.{name:name,status:status,ipFamily:kubernetesNetworkConfig.ipFamily}'
aws eks describe-cluster --name "$CLUSTER" --region "$REGION" \
  --query 'cluster.kubernetesNetworkConfig.serviceIpv4Cidr' --output text
aws eks update-kubeconfig --name "$CLUSTER" --region "$REGION"
kubectl config current-context
kubectl cluster-info
```

PLEASE PAUSE AND CHECK
If the account, cluster, context, region, or bucket is not the reviewed target, please pause and correct it before continuing. This reference currently supports EKS clusters where **ipFamily** is **ipv4**.

03 / PREFLIGHT

Prove capacity, access, and dependencies

Scheduler capacity is based on requested resources, not current utilization. IAM access and Kubernetes authorization are also separate checks.

Capacity and EBS

```
kubectl get nodes -L topology.kubernetes.io/zone
kubectl describe node <candidate-node>   # inspect Allocated resources

aws eks describe-addon --cluster-name "$CLUSTER" --region "$REGION" \
  --addon-name aws-ebs-csi-driver --query 'addon.{status:status,version:addonVersion}'
kubectl -n kube-system get deployment ebs-csi-controller
```

CAPACITY RULE
Each of three different eligible nodes needs 24 CPU and 50 GiB still available after existing pod and DaemonSet requests. Please use the node's **Allocated resources** view rather than **kubectl top**, because the scheduler works from requests.

Kubernetes authorization

```
kubectl auth can-i create namespaces
kubectl auth can-i create storageclasses.storage.k8s.io
kubectl auth can-i create networkpolicies.networking.k8s.io --all-namespaces
kubectl auth can-i create customresourcedefinitions.apiextensions.k8s.io
kubectl auth can-i create clusterroles.rbac.authorization.k8s.io
kubectl auth can-i create clusterrolebindings.rbac.authorization.k8s.io
```

NetworkPolicy enforcement

```
aws eks describe-addon --cluster-name "$CLUSTER" --region "$REGION" \
  --addon-name vpc-cni --query addon.configurationValues
kubectl api-resources | grep policyendpoints
```

Enforcement on	Enforcement off
Preferred for shared clusters. The EKS VPC CNI must have **enableNetworkPolicy: true**. Apply the Service-CIDR egress policy after the Restate namespace exists.	Formally supported only with an explicit trust decision. NetworkPolicy objects still appear, but every pod can reach the unauthenticated admin API and SDK endpoints.

04 / CHANGE PREPARATION

Prepare a reviewable installation

Run commands from the repository root. Keep substitutions visible in a working copy or branch, and compare every applied file with the approved change.

Order	Repository asset	Action
1	resources/00-namespaces.yaml	Create restate-operator and restate-apps; leave restate to the operator
2	resources/01-restate-snapshots-iam-policy.json	Render a temporary policy with the dedicated bucket
3	resources/02-restate-operator.values.yaml	Pass to Helm as values; this file is not a kubectl manifest
4	resources/03-gp3-storageclass.yaml	Create encrypted gp3 with Retain and delayed binding
5	resources/04-restate-cluster.yaml	Set region, bucket, and snapshot role ARN; then apply
6	resources/06-restate-service-cidr-egress.yaml	Render with the real Service CIDR where policy is enforced
Optional	resources/05-restate-compute.yaml	Apply only after a real SDK service image is set

Placeholder discipline

```
grep -RIn 'REPLACE_ME' resources
git diff -- resources
```

Placeholder	Required value
REPLACE_ME_SNAPSHOTS_BUCKET	Dedicated bucket name in IAM policy render and RestateCluster
REPLACE_ME_AWS_REGION	Target AWS region
REPLACE_ME_SNAPSHOTS_ROLE_ARN	arn:aws:iam:::role/-restate-snapshots
REPLACE_ME_SERVICE_CIDR	EKS serviceIpv4Cidr; render through stdin
REPLACE_ME_SERVICE_IMAGE	Only for the optional SDK-service example

OWNERSHIP BOUNDARY

Create **restate-operator** and **restate-apps**, and please leave **restate** for the operator to create and own from RestateCluster/restate. That ownership includes its StatefulSet, Services, ServiceAccount, PVCs, and policies.

05 / AWS PLANE

Create snapshot access with IRSA

The manual path assumes a pre-existing, dedicated S3 bucket. Verify its controls before creating a least-privilege policy and a role trusted only by the Restate ServiceAccount.

1. Apply repository-owned namespaces

```
kubectl apply -f resources/00-namespaces.yaml
kubectl get namespace restate-operator restate-apps
kubectl -n restate-apps get networkpolicy
```

2. Verify the bucket and OIDC provider

```
aws s3api head-bucket --bucket "$BUCKET"
aws s3api get-bucket-location --bucket "$BUCKET"
aws s3api get-public-access-block --bucket "$BUCKET"
aws s3api get-bucket-policy --bucket "$BUCKET" --query Policy --output text | jq

eksctl utils associate-iam-oidc-provider \
  --cluster "$CLUSTER" --region "$REGION" --approve
```

3. Create the IAM policy and role

```
sed "s/REPLACE_ME_SNAPSHOTS_BUCKET/$BUCKET/" \
  resources/01-restate-snapshots-iam-policy.json > /tmp/restate-snapshot-policy.json

aws iam create-policy --policy-name "${CLUSTER}-restate-snapshots" \
  --policy-document file:///tmp/restate-snapshot-policy.json

eksctl create iamserviceaccount --cluster "$CLUSTER" --region "$REGION" \
  --namespace restate --name restate --role-name "${CLUSTER}-restate-snapshots" \
  --attach-policy-arn "arn:aws:iam::$ACCOUNT:policy/${CLUSTER}-restate-snapshots" \
  --role-only --approve
```

VERIFY BEFORE CONTINUING

The role trust should name **system:serviceaccount:restate:restate**, and the policy and Restate snapshot destination should name the same bucket. The derived role name limits **CLUSTER** to 46 characters.

06 / CONTROL PLANE

Install and verify the operator

Helm installs the controller, CRDs, and cluster RBAC. Keep the chart version pinned and wait for the deployment rather than racing ahead to custom resources.

Install

```
helm upgrade --install restate-operator \
  oci://ghcr.io/restatedev/restate-operator-helm \
  --version 3.0.1 \
  --namespace restate-operator \
  --values resources/02-restate-operator.values.yaml \
  --wait --timeout 5m
```

Verify

```
kubectl -n restate-operator get deployment,pods
kubectl get crd \
  restateclusters.restate.dev \
  restatedeployments.restate.dev \
  restatecloudenvironments.restate.dev
```

Check	Expected	If it fails
Helm release	deployed in restate-operator	Inspect Helm status and controller events
Controller	Deployment Available; pod Ready	Read controller logs before creating RestateCluster
CRDs	All three definitions served	Please wait for the schemas before applying custom resources
Registry pull	Public anonymous pull succeeds	Clear stale ghcr.io credentials; authenticate only if your environment requires it

Registry diagnostic

```
helm registry logout ghcr.io || true
docker logout ghcr.io || true
helm pull oci://ghcr.io/restatedev/restate-operator-helm \
  --version 3.0.1 --destination /tmp
```

CRD LIFECYCLE
The chart marks its CRDs to survive Helm uninstall. That is intentional. During teardown, retain them until a cluster-wide check confirms that no Restate operator installation or custom resource remains.

07 / DATA PLANE

Create storage and bootstrap Restate

Apply storage first, confirm its safety settings, then apply the RestateCluster. The operator performs the one-time cluster provisioning after the pods form metadata membership.

1. Create and inspect the StorageClass

```
kubectl apply -f resources/03-gp3-storageclass.yaml
kubectl get storageclass restate-gp3 -o yaml
```

EXPECTED STORAGE

Each pod starts with 256 GiB. The StorageClass uses EBS CSI, encrypted XFS, 6000 IOPS, 500 MiB/s, WaitForFirstConsumer, allowVolumeExpansion, and reclaimPolicy Retain. Monitor /restate-data on every pod and expand with operational headroom. Retain preserves EBS after PVC deletion; it does not perform automatic recovery.

2. Apply the cluster and watch bootstrap

```
grep -n 'REPLACE_ME' resources/04-restate-cluster.yaml
kubectl apply -f resources/04-restate-cluster.yaml

kubectl get restatecluster restate -w
# second shell
kubectl -n restate get pods,pvc -w
```

3. Wait for the acceptance state

```
kubectl wait --for=condition=Ready restatecluster/restate --timeout=15m
kubectl get restatecluster restate -o jsonpath='{.status.provisioned}{"\n"}'
kubectl -n restate get pods -o wide
kubectl -n restate exec restate-0 -- restatectl status
```

KEEP PROVISIONING OPERATOR-MANAGED

Please leave provisioning to the operator while **spec.cluster.autoProvision** is enabled. Running **restatectl provision** at the same time can race the operator and split cluster initialization. If pods run but remain unready, inspect RestateCluster conditions and operator logs.

08 / BOUNDARIES AND BACKUP

Complete the network policy and verify snapshots

Where EKS VPC CNI enforcement is enabled, Restate needs an additional egress rule to the cluster Service CIDR on port 9080. Then prove snapshot access end to end instead of waiting for the automatic cadence.

Apply the Service-CIDR policy

```
SERVICE_CIDR="$(aws eks describe-cluster --name "$CLUSTER" --region "$REGION" \
  --query 'cluster.kubernetesNetworkConfig.serviceIpv4Cidr' --output text)"
echo "$SERVICE_CIDR"

sed "s|REPLACE_ME_SERVICE_CIDR|$SERVICE_CIDR|" \
  resources/06-restate-service-cidr-egress.yaml | kubectl apply -f -
kubectl -n restate get networkpolicy allow-egress-to-service-cidr
kubectl -n restate get policyendpoints
```

Traffic	Port	Intended boundary
Client -> Restate ingress	8080	Expose separately only when the customer designs ingress and authentication
Human -> admin	9070	Use kubectl port-forward and keep the admin endpoint private
Restate -> Restate	5122	Operator-managed node and metadata traffic
Restate -> SDK revision	9080	Operator labels plus Service-CIDR policy where pre-DNAT enforcement applies
Other workloads -> SDK	9080	Denied by the restate-apps ingress policy where enforcement is active

Trigger and verify a snapshot

```
kubectl -n restate exec restate-0 -- restatectl snapshots create-snapshot
aws s3 ls "s3://$BUCKET/restate/snapshots/" --recursive | head
```

ACCEPTANCE GATE
Before handoff, please confirm that the manual snapshot succeeds and objects appear in S3. This single test exercises the ServiceAccount annotation, OIDC trust, IAM policy, region, egress path, bucket, and Restate configuration.

09 / ACCEPTANCE

Validate and hand over

Capture evidence, not only a verbal success. The platform owner should be able to repeat these checks without reconstructing the installation session.

Five-minute health check

```
kubectl get restatecluster restate -o wide
kubectl get restatecluster restate \
  -o jsonpath='{range .status.conditions[*]}{.type}{"="}{.status}{"  "}{.message}{"\n"}{end}'
kubectl -n restate get pods -o wide
kubectl -n restate get pvc
kubectl -n restate get svc,networkpolicy
kubectl -n restate exec restate-0 -- restatectl status
```

Evidence to retain	Acceptance criterion
RestateCluster	status.provisioned=true and Ready=True
Placement	restate-0, restate-1, and restate-2 Ready on different nodes
Persistent data	Three PVCs Bound through restate-gp3; PV/PVC/AZ/EBS mapping recorded
Restate health	restatectl status reports healthy nodes, logs, and partitions
Snapshots	Manual snapshot command and S3 object listing captured
Exposure	svc/restate remains ClusterIP; port 9070 has no public route
Network	CNI enforcement state recorded; policies and policy endpoints inspected
Configuration	No active REPLACE_ME value remains in an applied payload

Operator access

```
kubectl -n restate port-forward svc/restate 8080:8080 9070:9070
# in a second shell
curl --fail --silent localhost:9070/services | jq
```

HANDOFF RECORD
Record the AWS account and region, EKS cluster and context, namespaces, bucket, role and policy ARNs, image and chart versions, capacity decision, NetworkPolicy enforcement state, snapshot evidence, ingress owner, and data-retention owner.

10 / OPTIONAL APPLICATION STEP

Optional: add an SDK service when useful

The infrastructure installation is complete without an application. If the customer also wants an SDK-service proof, use a real image and treat the operator-managed revision lifecycle as different from a normal Deployment rollout.

Apply the example skeleton

```
# Set REPLACE_ME_SERVICE_IMAGE to an image that listens on port 9080.
grep -n 'REPLACE_ME' resources/05-restate-compute.yaml
kubectl apply -f resources/05-restate-compute.yaml
kubectl -n restate-apps get restatedeployments,pods -w
```

Requirement	Why
Container listens on 9080	Restate invokes the SDK endpoint on this port
Port is named restate	The operator uses the named port to construct the registration URL
spec.restate.register.cluster is restate	The revision registers with the installed Restate cluster
RestateDeployment becomes Ready	Confirms image pull, pod health, registration, admin access, and the 9080 path
Old revisions are allowed to drain	Pinned in-flight invocations may still require the previous revision

Normal rollout
Change the pod template, review the diff, and apply. The operator creates a versioned ReplicaSet and Service, registers the new revision, and drains the old revision.

Rollback
Reapply a previously known-good pod template. Please allow the old ReplicaSet to drain normally, because it may still own in-flight work.

NETWORK DIAGNOSTIC
Ready SDK pods with a RestateDeployment that remains NotReady often indicate a missing Service-CIDR egress policy. From a Restate pod, compare access to the revision pod IP and its Service ClusterIP on port 9080.

Customer-facing exposure

Keep the operator-managed **svc/restate** as a ClusterIP because it carries both ingress 8080 and unauthenticated admin 9070. If shared ingress is required, create a separately owned Service or Ingress for **8080 only**, prefer internal exposure, add authentication, update the allowed network peer, and set the advertised ingress address deliberately.

11 / TROUBLESHOOTING AND DATA SAFETY

Diagnose by layer; preserve recovery options

To keep troubleshooting predictable, finish investigating the current change before making another. A helpful sequence is scheduling and storage, operator reconciliation, Restate status, IAM, and then network paths.

Symptom	First evidence	Common cause or safe action
Pod Pending	Pod describe, events, node Allocated resources	Capacity, anti-affinity, taint, EBS CSI, or an unavailable AZ
Pods Running, not Ready	RestateCluster YAML, Restate logs, operator logs	Provisioning, peer DNS, or config validation; keep provisioning operator-managed
Snapshot fails	ServiceAccount, role trust, pod env, S3 path	Wrong ARN, bucket, region, IAM policy, OIDC, or private endpoint blocked by egress
SDK revision not Ready	RestateDeployment describe, operator logs, pod IP vs ClusterIP	Image/port issue or missing Service-CIDR egress rule
Admin unreachable	Service type, port-forward output, policy state	Use svc/restate through kubectl port-forward and keep 9070 private

Before removal or other data-affecting changes

```
kubectl -n restate get pvc -o wide
kubectl get pv \
  -o jsonpath='{range .items[*]}{.metadata.name}{"  "}'\
'{.spec.claimRef.namespace}/{.spec.claimRef.name}{"  "}'\
'{.spec.csi.volumeHandle}{"\n"}{end}'
kubectl -n restate exec restate-0 -- restatectl snapshots create-snapshot
aws s3 ls "s3://$BUCKET/restate/snapshots/" --recursive | head
```

RECOVERY BOUNDARY
Retained EBS PVs and S3 snapshots protect different failure modes, but neither is an automatic disaster-recovery workflow. Released PVs keep their old claim references and do not bind to replacement PVCs automatically.

Safe-change pattern

- Verify cluster health and a recent snapshot before changing runtime sizing, storage, image, chart, or experimental settings.
- Review one change at a time. Pod-template changes can roll all three stateful pods and move partition leadership.
- Monitor /restate-data on every pod. Expand with operational headroom, then confirm capacity.
- For upgrades, revalidate release compatibility and profile-derived settings; changing only the image is incomplete.

12 / TEARDOWN AND SOURCE MAP

A safe order for removal

A manual deployment can be removed safely, but data and cluster-scoped definitions require explicit decisions. This is a boundary summary, not authorization to delete production data.

Order	Action	Safety note
1	Pause new traffic; snapshot; capture PV/PVC/AZ/EBS mapping	Complete before deleting a Kubernetes parent
2	Delete every RestateDeployment and wait for finalizers	Allow finalizers to finish the drain
3	Delete RestateCluster/restate and wait for namespace deletion	PVCs disappear; Retain should leave EBS volumes
4	Record Released PV and EBS mapping again	Preserves the link between Kubernetes and billable AWS volumes
5	Uninstall operator; then namespaces and StorageClass	Retain CRDs until the cluster-wide dependency check
6	Decide retention for EBS, S3, IAM, and OIDC	OIDC is shared; bucket and volumes may still contain recovery data

WHAT SURVIVES
With the repository defaults, three EBS volumes, the S3 bucket and objects, IAM role and policy, and the cluster OIDC provider can remain after Kubernetes cleanup. The EKS cluster also remains because this repository leaves it unchanged.

Source map

Need	Repository source
Product and ownership overview	README.md
Full deployment gates	docs/01-prerequisites.md
Canonical manual commands	docs/02-runbook.md
Traffic, bootstrap, IAM, and invariants	docs/00-architecture.md
Health, diagnosis, recovery, and teardown	docs/05-operations.md
SDK-service rollout and draining	docs/03-deploying-services.md
Canonical applied assets	resources/00 through resources/06

Reference baseline

This companion was generated from repository commit **b31a2e8274bf56fc7493188b903e55106ac42d08** dated 2026-09-04. Commands and versions should be revalidated if used with a different revision.

Official background: https://docs.restate.dev/foundations/key-concepts
Operator project: https://github.com/restatedev/restate-operator

Source-of-truth note: If this PDF and the checked-out repository disagree, please pause and use the repository revision under change control. Re-run **nix-shell --run ./scripts/validate.sh** before deployment.